Pipedrive Integration

See also: [TESTING_GUIDE.md]

1. Overview

1.1 Description

Pipedrive Integration connects **Pipedrive** (sales CRM platform) with **Newo Platform**.

It enables:

- Checking real-time appointment availability by querying Pipedrive Activities
- Booking appointments with automatic contact (person) creation
- Cancelling existing appointments
- Existing customer detection with upcoming booking retrieval
- Automatic customer context injection into agent prompt

This integration is designed for **sales teams and service businesses** who need **automated appointment scheduling and contact management through a conversational AI agent** (voice or chat).

1.2 Use Cases

- When a client asks about available slots → Query Pipedrive Activities and calculate free time slots
- When a client wants to book an appointment → Find or create person, then create Activity in Pipedrive
- When a client wants to cancel an appointment → Delete the Activity from Pipedrive
- When a caller's phone is identified → Look up existing person and inject their context + upcoming bookings into prompt
- When a conversation starts → Optionally preload nearest available slots silently

1.3 Supported Features

Feature	Supported	Notes
Check Availability	✅	Calculates free slots from existing Activities
Book Appointment	✅	Creates Activity with auto person creation
Cancel Appointment	✅	Deletes Activity from Pipedrive
Existing Client Detection	✅	By phone, with prompt context injection
Upcoming Booking Retrieval	✅	Future Activities stored in persona
Silent Availability Preload	✅	Optional check on conversation start
Configurable Slot Duration	✅	Default 30 min
Configurable Booking Window	✅	Days ahead to show (default: 1)
Timezone Handling	✅	Converts between local and UTC
Reschedule Appointment	❌	Cancel + rebook as workaround
Deal/Pipeline Management	❌	Not implemented

Multi-Location Support	❌	Single Pipedrive instance per integration

2. Requirements

Before installation:

- Active **Pipedrive** account
- **Pipedrive API Token** (from Personal Preferences → API)
- Pipedrive account URL (e.g., `https://yourcompany.pipedrive.com`)

3. How to Setup

3.1 Step 1 — Obtain API Token from Pipedrive

1. Log in to your **Pipedrive** account
2. Navigate to **Personal Preferences → API**
3. Copy your **Personal API Token**
4. Store it securely

3.2 Step 2 — Connect in Platform

1. Open **Newo → Projects**
2. Set the following attributes in **Builder / Attributes**:

Attribute	Required	Description
`pipedrive_api_token`	✅	API token for authentication (`x-api-token` header)
`pipedrive_base_api_url`	❌	Pipedrive account URL (default: `https://newo.pipedrive.com`)
`pipedrive_duration`	❌	Appointment duration in minutes (default: `30`)
`pipedrive_show_for_days`	❌	Days ahead for availability (default: `1`)
`pipedrive_slots_available`	❌	Enable availability checking (default: `True`)
`pipedrive_booking_available`	❌	Enable booking capability (default: `True`)
`pipedrive_cancelation_available`	❌	Enable cancellation capability (default: `True`)
`pipedrive_check_availability_on_conversation_start`	❌	Pre-load slots on session start (default: `False`)

3. Click **Save + Publish All**
4. On publish, the SetupFlow automatically:
    - Validates the API token
    - Creates the HTTP connector for Pipedrive

- Injects booking and availability payload schemas for the agent

4. How to Use

This section explains how the integration works, how to configure automation, and how to test it.

4.1 How the Integration Works

The integration uses Pipedrive **Activities** as appointments and **Persons** as contacts.

- A **Trigger** is a system event that starts a flow
- An **Action** is the API operation performed in Pipedrive

Available Triggers

Trigger	Event	Description
Check Availability	`get_available_slots`	When the agent needs to look up open slots
Book Appointment	`book_slot`	When the agent confirms a booking
Cancel Appointment	`convo_agent_cancel_booking`	When the agent processes a cancellation
Phone Identified	`define_user_phone_number`	When the caller's phone is detected
Session Start	`session_started`	Optional silent availability preload
Setup	`project_publish_finish`	Initializes integration on deploy

Available Actions

- **Search Person** — Find contact by phone (GET `/v1/persons/search`)
- **Create Person** — New contact with name and phone (POST `/api/v2/persons`)
- **Get Activities** — Fetch booked appointments (GET `/api/v1/activities`)
- **Get Person Activities** — Fetch contact's bookings (GET `/api/v2/activities?person_id={id}`)
- **Create Activity** — Book an appointment (POST `/v1/activities`)
- **Delete Activity** — Cancel an appointment (DELETE `/api/v2/activities/{id}`)

4.2 Architecture Diagrams

Overall Sequence

```
sequenceDiagram
    participant U as User
    participant A as ConvoAgent
    participant I as Pipedrive Integration
    participant API as Pipedrive API

    Note over I: Setup (on publish)
    A->>I: project_publish_finish
    I->>I: Create connector, inject schemas
    I->>A: Tools configured

    Note over U,API: Existing Client Detection
    A->>I: define_user_phone_number (phone)
```

```
    I->>API: GET /v1/persons/search (phone)
    API-->>I: Person data
    I->>A: Inject ExistingCustomerInfo into prompt
    I->>API: GET /api/v2/activities?person_id=(id)
    API-->>I: Future bookings
    I->>A: Store bookings in persona

    Note over U,API: Availability Check
    U->>A: "Any slots tomorrow?"
    A->>I: get_available_slots (date, time)
    I->>API: GET /api/v1/activities (date range, done=false)
    API-->>I: Booked activities
    I->>I: Calculate free slots (subtract booked, slice by duration)
    I->>A: result_success (availability[])
    A->>U: "Here are the available times..."

    Note over U,API: Booking
    U->>A: "Book 2:00 PM please"
    A->>I: book_slot (customerInfo, dateTime)
    I->>API: GET /v1/persons/search (phone)
    alt Person not found
        I->>API: POST /api/v2/persons (name, phone)
        API-->>I: person_id
    end
    I->>API: POST /v1/activities (due_date, due_time, duration, person_id)
    API-->>I: activity_id
    I->>A: result_success (booking_id)
    A->>U: "Your appointment is confirmed!"

    Note over U,API: Cancellation
    U->>A: "Cancel my appointment"
    A->>I: convo_agent_cancel_booking (booking_id)
    I->>API: DELETE /api/v2/activities/(booking_id)
    API-->>I: Confirmation
    I->>A: result_success (cancelled)
    A->>U: "Your appointment has been cancelled"
```

AvailabilityFlow

```
flowchart TD
    E["get_available_slots<br>or session_started"] --> GATE{"Slot
check<br>enabled?"}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| RANGE["Calculate date range:<br>date → date + show_for_days"]
    RANGE --> API["GET /api/v1/activities<br>{updated_since, updated_until,
<br>done=false}"]
    API --> OK{"HTTP 200?"}
    OK -->|"No"| ERR["ResultError"]
    OK -->|"Yes"| CALC["Parse booked activities<br>Create duration-based
slots<br>Subtract overlapping"]
```

```
    CALC --> TZ["Convert UTC<br>to local timezone"]
    TZ --> OUT["result_success<br>{availability[]}"]
```

BookingFlow

```
flowchart TD
    E["book_slot"] --> GATE{"Booking<br>enabled?"}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| SEARCH["GET /v1/persons/search<br>{phone, fields: phone}"]
    SEARCH --> FOUND{"Person<br>found?"}
    FOUND -->|"Yes"| PID["Use person_id"]
    FOUND -->|"No"| CREATE["POST /api/v2/persons<br>{name, phones}"]
    CREATE --> PID
    PID --> UTC["Convert time to UTC"]
    UTC --> API["POST /v1/activities<br>{due_date, due_time, duration,<br>subject,
type: meeting,<br>person_id}"]
    API --> OK{"HTTP 200?"}
    OK -->|"No"| ERR["ResultError"]
    OK -->|"Yes"| OUT["result_success<br>{booking_id: activity_id}"]
```

CancellationFlow

```
flowchart TD
    E["convo_agent_cancel_booking"] --> GATE{"Cancellation<br>enabled?"}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| VAL{"booking_id<br>present?"}
    VAL -->|"No"| ERR["ResultError"]
    VAL -->|"Yes"| API["DELETE /api/v2/activities<br>/{booking_id}"]
    API --> OK{"Success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| OUT["result_success<br>{cancel_booking}"]
```

ExistingClientFlow

```
flowchart TD
    E["define_user_phone_number"] --> SEARCH["GET /v1/persons/search<br>{phone}"]
    SEARCH --> FOUND{"Person<br>found?"}
    FOUND -->|"No"| DONE["No action"]
    FOUND -->|"Yes"| INJECT["Inject ExistingCustomerInfo<br>into agent prompt<br>
{name, email}"]
    INJECT --> APPTS["GET /api/v2/activities<br>?person_id={id}"]
    APPTS --> FILTER["Filter future<br>activities only"]
    FILTER --> SAVE["Store in persona<br>attribute: bookings<br>(deduplicated by
booking_id)"]
```

4.3 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice channel). BookingFlow and CancellationFlow have dedicated test skills for connectivity validation.

4.4 Testing and Verification

To test the integration:

1. **Setup**: Publish the project and verify schemas are injected
2. **Existing Client**: Start a conversation from a known phone — verify customer info injected into prompt
3. **Availability**: Ask "What slots are available tomorrow?" — verify free slots calculated correctly
4. **Booking**: Complete a booking — verify Activity created in Pipedrive with correct person link
5. **Cancellation**: Cancel a booking — verify Activity deleted from Pipedrive

If no action occurs:

- Ensure `pipedrive_api_token` is set correctly
- Verify `pipedrive_base_api_url` points to your Pipedrive account (not the default)
- Confirm the integration was re-published after changes
- Check feature flags are enabled (`pipedrive_slots_available` , `pipedrive_booking_available` , `pipedrive_cancelation_available`)
- Verify timezone is set correctly via `pipedrive_timezone`

> *Note: This integration performs real operations in Pipedrive. Activities and persons created or deleted through the agent are reflected in the live CRM system.*

5. FAQ

Q: Does the integration import historical data? A: No. Only Activities created after activation are managed. Existing client detection retrieves future Activities for context only.

Q: Can I connect multiple Pipedrive accounts? A: Each integration instance supports one API token. For multiple accounts, create separate integration instances.

Q: How does availability checking work? A: The integration fetches incomplete Activities (done=false) within the date range and calculates free time slots by finding gaps between booked activities. Slot duration is configurable via `pipedrive_duration` (default: 30 minutes).

Q: What happens if a contact already exists? A: The integration searches persons by phone number. If found, the existing person is used for booking. If not, a new person is created with name and phone.

Q: Why is the default booking window only 1 day? A: `pipedrive_show_for_days` defaults to 1 for performance. Increase it to show more days of availability (e.g., 7 for a week).

Q: How are phone numbers handled? A: Phone numbers are expected in E.164 format. The leading `+` is stripped for Pipedrive search queries. Search matches against all phone numbers associated with a person.

Q: What Pipedrive entity represents an appointment? A: Appointments are stored as Pipedrive **Activities** of type "meeting". The Activity subject includes the customer name and formatted time (e.g., "John Doe — Sep 28 at 2:00 PM").